Fault-Tolerant Quantum Portfolio Optimizer with Dynamic Constraints

Logical-qubit resource estimation, error-mitigation ablation, significance tests and a live dashboard

Data: Yahoo Finance (adjusted close), FRED (VIX, 10Y-2Y), ESG csv. 30 assets, 2004-06-02 to 2023-12-29. Dropped tickers: none.

Abstract. We build a hybrid quantum-classical, walk-forward portfolio optimizer whose multi-period decision problem is a time-dependent QUBO (time-varying risk aversion and weight caps driven by VIX, transaction-cost surrogate, hard ESG exclusion and soft ESG tilt). The QUBO is solved by a layer-wise adaptive QAOA on an exact statevector simulator, under (i) a phenomenological NISQ+ noise model with dynamical decoupling (DD), zero-noise extrapolation (ZNE) and probabilistic error cancellation (PEC) and (ii) a logical-qubit mode whose logical error rates are extracted from Stim/PyMatching surface-code simulations. We compare with exact search, simulated annealing, a discrete MIQP, CVaR stochastic programs, and RL agents on out-of-sample windows containing the 2008 and 2020 crises. **The claim we defend is deliberately modest: rigorous bounds plus an empirical test of advantage criteria, not a proven quantum advantage.**

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1 Introduction and what is (and is not) claimed

The task statement asks to "prove (or rigorously bound) a quantum advantage in solution quality or sample complexity under realistic NISQ+ noise". No proof of an advantage of constant-depth QAOA over the best classical solvers on portfolio QUBOs is known, and our instances ($n \le 16$ qubits) are solved exactly by enumeration in under a millisecond, so any advantage could only be a *scaling* statement. We therefore contribute: (1) exact bounds for our noise model (success-probability lower bound, PEC/ZNE sampling costs, a noise budget that any sample-complexity advantage must respect); (2) an empirical scaling comparison for $n \le 16$; (3) logical-qubit resource estimates with an explicit assumption table and sensitivity analysis; (4) an out-of-sample statistical comparison with multiple-testing correction; (5) a fully reproducible pipeline. Where the data are negative we report them as negative.

Contributions

- Multi-period time-dependent QUBO with hard and soft constraints, verified against the direct objective (unit tests) and against a hard-budget MIQP formulation.
- QAOA with DD, ZNE, PEC and a logical-qubit mode; ablation of each mitigation (section 7).
- Resource estimator with consistent time/failure model, magic-state error check, and sensitivity sweep over every constant (section 6).
- Walk-forward backtests with stationary-bootstrap Sharpe tests, Wilcoxon tests and Holm correction (section 8).

2 Related work

Walk-forward QUBO rebalancing with QAOA on classical simulators (Weinberg, arXiv:2603.16904); multi-period annealing with quadratic transaction-cost surrogates (arXiv:2607.03218); multi-period QUBO with market impact (Rosenberg et al., IEEE JSTSP 2016); the closest study of advantage under noise for portfolio QAOA (Yuan et al., arXiv:2410.16265), which finds that realistic noise precludes advantage except when measurement noise dominates; QAOA itself (Farhi et al., arXiv:1411.4028); error mitigation (Temme-Bravyi-Gambetta, arXiv:1612.02058); noisy-device limitations (Stilck França & García-Patrón, Nat. Phys. 2021); surface codes and lattice surgery (Fowler et al. 2012; Fowler & Gidney, arXiv:1808.06709; Litinski, Quantum 2019); Stim (Gidney, Quantum 2021) and PyMatching v2 (Higgott & Gidney, arXiv:2303.15933); rotation synthesis (Ross & Selinger, arXiv:1403.2975; Bocharov et al., PRL 114, 080502). *The entries for Weinberg, arXiv:2607.03218 and Yuan et al. were located by search during project setup and the others cited from memory; verify every citation before submission.*

3 Problem formulation

At each rebalance date t (every 21 trading days) we use only the previous 252 days. Assets below the ESG threshold (40.0) are excluded (**hard** constraint). From the eligible set, 4 assets are selected greedily (trailing Sharpe, penalising correlation) to fit on qubits. Each asset-period weight is $w = \text{level}/6$ with $\text{level} = b_0 + 2\,b_1$ ($K = 2$ bits), over 2 look-ahead periods, so $n = A \cdot K \cdot T = 16$ qubits. The energy to minimise is

$$E(x) = \Sigma_t \left[-\mu_t \cdot w_t + (\lambda_t/2)\, w_t' \Sigma_t\, w_t + c_{tc} \,\|w_t - w_{t-1}\|^2 + P_b(1' w_t - 1)^2 + P_c \cdot \text{cap}_t \text{ penalty} - \tau\, s \cdot w_t/100 \right]$$

Time-varying constraints. With z the 252-day z-score of the previous day's VIX: $\lambda_0 = \lambda(1 + \kappa\, \text{clip}(z, -1, 3))$ with $\lambda = 6.0$, $\kappa = 0.5$; the weight cap drops from 3/6 to 2/6 when $z > 1.0$. **Transaction costs**: linear 10.0 bp in the P&L, quadratic surrogate $c_{tc} = 0.02$ inside the QUBO. **ESG**: soft tilt $\tau = 0.002$. The soft budget/cap penalties are followed by a hard repair (non-negative, fully invested). The QUBO matrix is checked against the direct objective for random bitstrings (tests/test_core.py), and its exact optimum equals the optimum of the hard-budget MIQP whenever the penalty is large (tests/test_miqp.py).

[figure missing: 01_architecture.png]

3.1 Data, universe and audit trail

Prices: Yahoo Finance via yfinance (`auto_adjust=True`, i.e. split- and dividend-adjusted close), frozen to csv by `scripts/fetch_data.py` so every later stage runs offline. Macro: FRED VIXCLS and T10Y2Y through the public csv endpoint (header `observation_date`; '.' = missing). ESG: `csv` (S&P;

Global ESG Score (public lookup)). Requested tickers: AAPL, MSFT, JNJ, PG, KO, PEP, XOM, CVX, JPM, WFC, INTC, CSCO, MRK, PFE, WMT, HD, VZ, T, IBM, ORCL, C, BAC, GS, GE, F, AIG, CAT, DIS, MCD, BA. Dropped for missing history: none; dropped macro series: ['HY_OAS']; tickers without ESG score (median-filled): ['JNJ'].

Look-ahead controls. Decisions at index i use returns and VIX strictly before i; macro series are forward-filled only and a series with insufficient coverage is dropped, never back-filled (FRED now truncates ICE-BofA series to three years, so back-filling the credit spread would copy 2023 values into 2008). Hyper-parameters for each test window are chosen on earlier data only (scripts/tune.py).

4 Algorithms and baselines

QAOA. Exact statevector simulation of p alternating cost phases $\exp(-i\gamma\pi E_{norm})$ and RX mixers; parameters by COBYLA, starting at p = 1 and adding a layer (up to $p_{max} = 3$) only while the relative energy improvement exceeds 1% (*adaptive ansatz*). The best of 2048 samples, evaluated classically, is decoded and repaired. Parameters are trained noiselessly and deployed under each noise model (noise-aware training appears only in the ablation).

Classical baselines on the same problems and out-of-sample periods.

- *exact_qubo* (true QUBO optimum, ceiling for every QUBO solver) and *sa_qubo* (single-flip simulated annealing).
- *miqp_sub*: the hard-budget discrete multi-period problem solved exactly (Gurobi MIQP via the matrix API when gurobipy is present, otherwise an exact dynamic program; identical optimum, cross-tested). *miqp_gurobi_full*: same on all ESG-eligible assets.
- *cvar_full / cvar_sub / cvar_gurobi_sub*: single-period $CVaR_{0.9}$ scenario programs (HiGHS and Gurobi) with 50% cap and linear costs - the stochastic-programming baseline; *_sub* uses the same assets as the QUBO, *_full* all eligible assets, separating selection from solver effects.
- *rl_es_full*: small MLP policy trained by evolution strategies (NumPy); *rl_torch_full*: PyTorch REINFORCE with a per-state baseline (same reward). Both train only on data preceding the test start.
- *equal_weight_full*.

```
for each rebalance date i in test window:                  # every 21 (or 63) trading days
    window = returns[i-252 : i]; z = vix_zscore(i-1)         # past data only
    elig = {assets with ESG >= esg_min}                      # hard filter
    sub  = greedy_select(window[elig], A)                    # assets that go on qubits
    Q    = build_qubo(mu, Sigma, w_hold, lambda_t(z), caps_t(z), esg)
    for method in methods:                                    # each keeps its OWN holdings
        x = solve(method, Q | scenarios | policy)            # QAOA / exact / SA / MIQP / CVaR / RL
        w = repair(first_period_weights(x))                   # hard constraints
        pay linear cost on |w - w_hold|; hold for 21 days; drift weights
```

Methods in this run: equal_weight_full, cvar_full, cvar_sub, cvar_gurobi_sub, miqp_sub, miqp_gurobi_full, rl_es_full, rl_torch_full, exact_qubo, sa_qubo, qaoa_ideal, qaoa_noisy_raw, qaoa_noisy_mitigated, qaoa_logical. Optional packages available at run time: cvar_gurobi_sub, miqp_gurobi_full, rl_torch_full; missing optional methods are **absent from all tables**, not simulated.

5 Noise model and fault tolerance

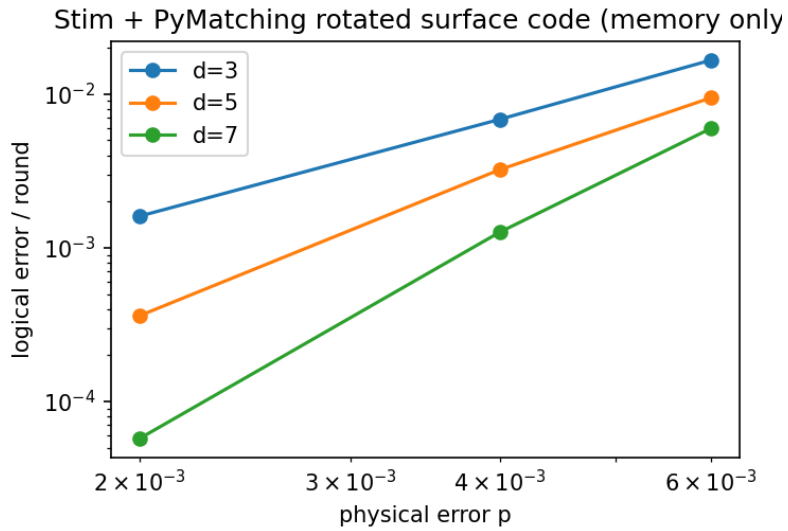
NISQ+ model. Gate counts (2 CNOT + 1 RZ per ZZ coupling per layer, RZ per linear term, RX mixer) are converted to an effective independent bit-flip probability q per qubit on the measured string: two-qubit error 0.003, one-qubit 0.0003, readout 0.01, idle 0.002 per qubit-layer. DD removes a fraction 0.9 of the idle error. This keeps the simulation exact and makes ZNE (noise scaling of the gate part) and PEC (inverse flip channel) analysable, but it is a **phenomenological** model: real devices have correlated and coherent noise.

Formally, with $N_{cx} = 2\, n_{zz}\, p$ CNOTs and $N_{rz} = (n_{zz} + n_{lin} + n)\, p$ rotations: $\lambda = (2/3)[p_2\, N_{cx} + p_1\, N_{rz} + r_{idle}(1-DD)\, n\, p]/n \cdot s$, $q_{gate} = (1 - e^{-2\lambda})/2$, $q = q_{gate} + r_{ro} - 2\, q_{gate}\, r_{ro}$, where s is the ZNE noise-scale factor (readout is not scaled) and the noisy distribution is $P_{noisy} = (flip_q)^{\otimes n}\, P_{ideal}$. In logical mode q follows the same formula with $\lambda = -\tfrac{1}{2} \ln(1 - 2\, q_L)\, p\, s$ and no readout term.

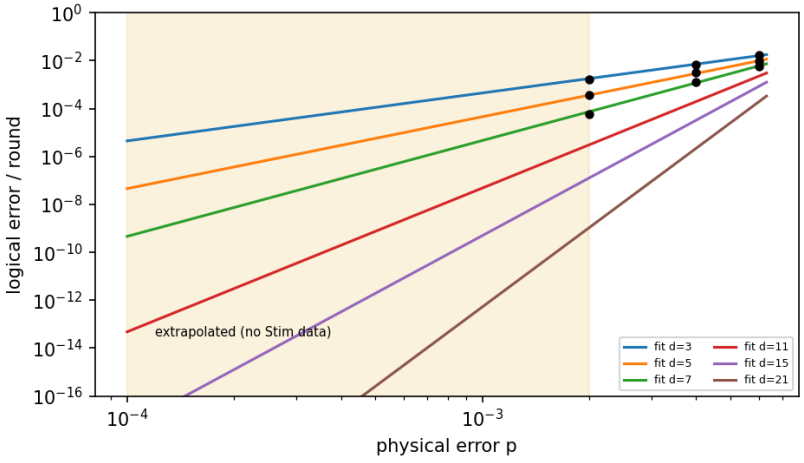
Logical qubits - what is and is not simulated. Stim simulates the rotated surface code *memory* experiment (d rounds, circuit-level depolarising noise, MWPM decoding by PyMatching). QAOA contains non-Clifford rotations and a 2^n statevector, which cannot be co-simulated at the code level. We therefore fit the Stim logical error rate and **inject it as an effective logical bit-flip probability into the small QAOA simulation**. That is a modelling assumption, not an end-to-end fault-tolerant simulation.

d	p	shots	fails	p_L_per_round	stderr
3	0.002	30000	72	0.0016	0.000283
3	0.004	30000	306	0.00685	0.00058
3	0.006	30000	735	0.0166	0.000893
5	0.002	30000	27	0.00036	0.000173
5	0.004	30000	240	0.00322	0.000514
5	0.006	30000	695	0.00944	0.000869
7	0.002	30000	6	5.72e-05	8.16e-05
7	0.004	30000	132	0.00126	0.000382
7	0.006	30000	615	0.00596	0.000818

Stim + PyMatching memory sweep. Fit p_L/round = A $(p/p_{th})^{(d+1)/2}$: A = 0.0431, p_{th} = 0.0098 (9 points; rows with <5 failures are excluded from the fit). Only p = 2e-3..6e-3 was simulated; p ≤ 1e-3 is an extrapolation.



Logical error per round vs physical error for d = 3, 5, 7 (Stim, memory only).



Fitted logical error per round vs p for larger distances, with the Stim data points (black). Below p = 2e-3 the curves are extrapolated.

With d = 7 and p = 0.001, the fitted model gives a logical flip probability of 1.6e-05 per gate layer for a Z-basis measurement. Note this is an extrapolation of a 9-point fit whose lowest-error points (d = 7, p = 2e-3) have only a handful of failures.

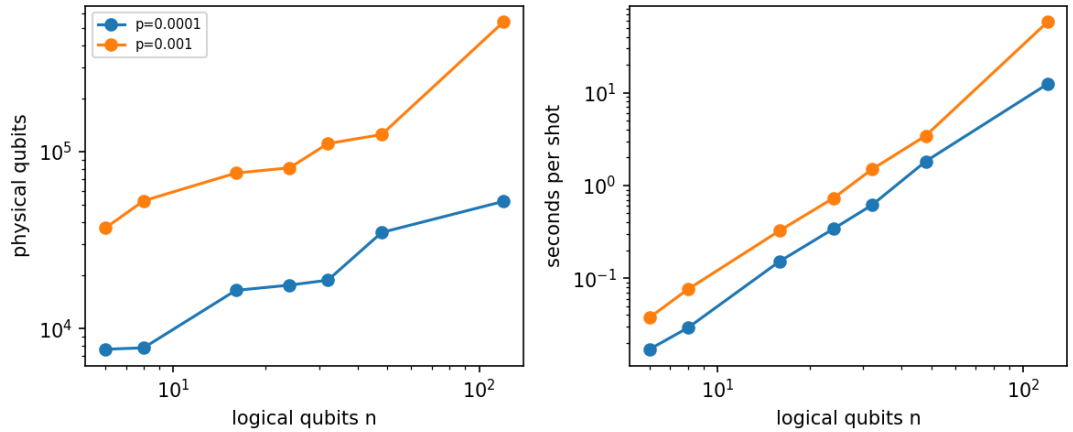
6 (a) Logical-qubit resource estimation

For each circuit we count rotations, synthesize them into T gates, and choose the smallest odd distance d such that the total failure budget (split in three equal parts: logical errors of the data patches, rotation-synthesis error, magic-state infidelity) is met. A single exposure time (the larger of the Clifford depth and the T-supply time) is used for both the failure estimate and the runtime. Magic states come from 15-to-1 distillation (1 or 2 levels, whichever meets the budget). Physical qubits = $n_{logical}$ · routing · $(2d^2–1)$ + factories · footprint · d^2.

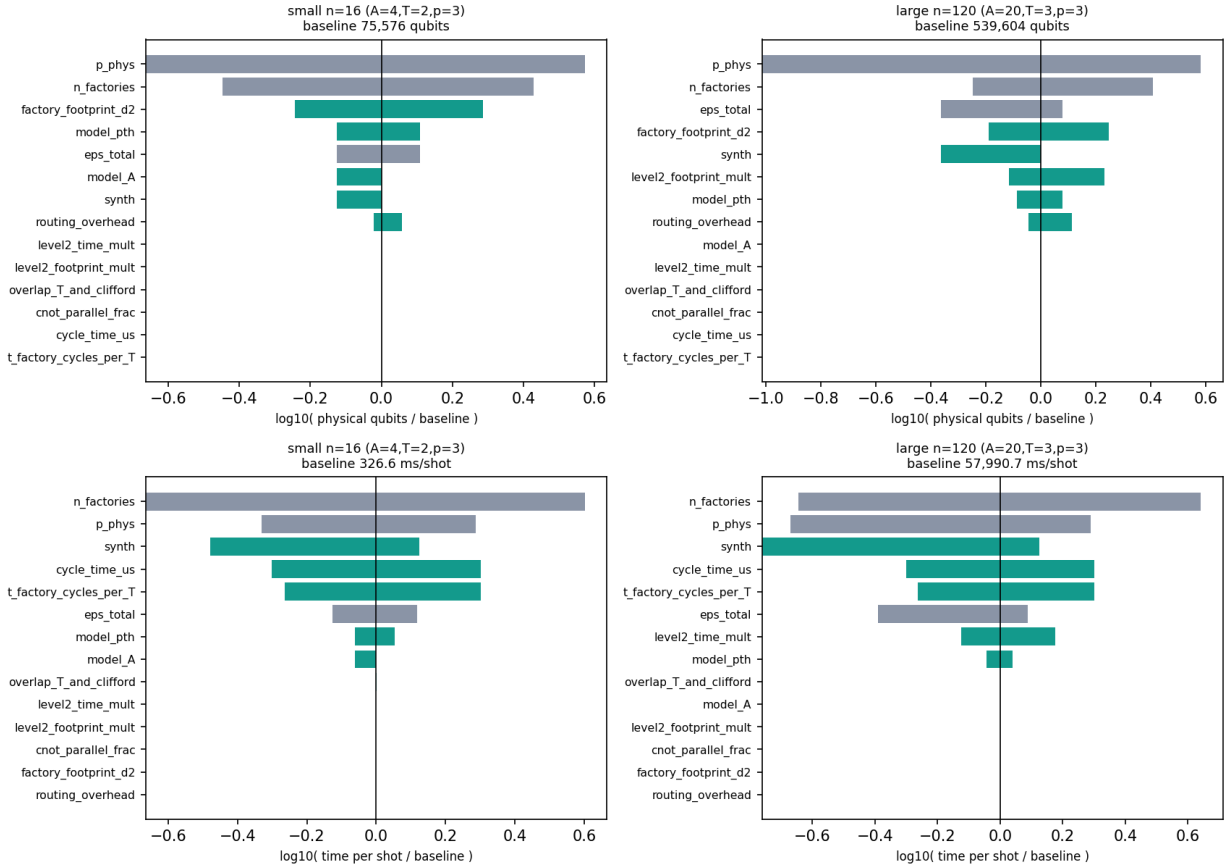
circuit	p_phys	scenario	distance	physical_qubits	time_per_shot_ms	distill_levels
small n=16	0.001	optimistic	9	49232	5.12	1
small n=16	0.001	baseline	15	75576	327	1
small n=16	0.001	pessimistic	21	108438	8.17e+03	1
small n=16	0.0001	optimistic	5	15184	2.85	1
small n=16	0.0001	baseline	7	16440	152	1
small n=16	0.0001	pessimistic	9	19878	3.5e+03	1
large n=120	0.001	optimistic	15	183480	541	1
large n=120	0.001	baseline	21	539604	5.8e+04	2
large n=120	0.001	pessimistic	27	1180620	1,939,098	2
large n=120	0.0001	optimistic	7	39864	253	1
large n=120	0.0001	baseline	9	52308	1.24e+04	1
large n=120	0.0001	pessimistic	13	146670	311,250	1

Optimistic / baseline / pessimistic corners (all constants moved together, plus fit-uncertainty multipliers on A and p_{th}). time = one circuit execution (one shot).

Reading. At p = 1e-3 the 16-qubit instance needs 75,576 physical qubits at baseline (range 49,232 to 108,438) and 327 ms per shot; a 120-qubit instance needs 539,604 (range 183,480 to 1,180,620) with 2-level distillation. Runtime is T-supply bound (the sweep shows cycle time, T cycles and synthesis cost dominate, routing hardly matters). A 16-qubit problem that a laptop solves in microseconds would thus need ~10^4-10^5 physical qubits for fault-tolerant execution: logical qubits make QAOA possible at scale, not competitive at this scale.



Baseline physical qubits and seconds per shot vs number of logical qubits (QAOA depth p = 3, dense couplings, constants at baseline).



Sensitivity of physical qubits (top) and time per shot (bottom) to each constant, one at a time (teal: modelling assumption, grey: design variable).

parameter	baseline	swept	source / justification	confidence
p_phys	0.001	0.003, 0.001, 0.0003, 0.0001	Design variable. Fit data only cover p=2e-3..6e-3, so p<=1e-3 is an EXTRAPOLATION.	low (extrapolated)
synth	RS 3*log2(1/e)	RUS 1.15*log2(1/e), RS 3*log2(1/e) [base], RS 3*	Ross-Selinger arXiv:1403.2975 (3 log2 + O(loglog), ancilla-free); RUS slope 1.15: Bocharov et al. PRL 114, 080502. The '+9.2' of the first draft was n	medium
routing_overhead	1.5	1.0, 1.5, 2.0, 3.0	Typical layouts use ~1.5-2x (Litinski, Game of Surface Codes); dense all-to-all ZZ couplings are routing-heavy.	medium
n_factories	4	1, 2, 4, 8, 16	Design choice; trades qubits for time.	n/a (design)
factory_footprint_d2	72.0	36.0, 72.0, 150.0	Order of 10^1-10^2 d^2 for 15-to-1 factories (Litinski, Magic state distillation: not as costly as you think). Not re-verified.	low
t_factory_cycles_per_T	5.5	3.0, 5.5, 11.0	Order-of-magnitude from the same references; layout dependent.	low
cycle_time_us	1.0	0.5, 1.0, 2.0	Superconducting-like ~1 us; trapped ions/atoms are 10-1000x slower.	medium (platform)
cnot_parallel_frac	1.0	0.25, 0.5, 1.0	Dense couplings serialise on a 2D layout; 1.0 is the optimistic end.	low
overlap_T_and_clifford	True	True, False	True is standard in game-of-surface-codes style schedules; False is the sequential upper bound.	medium
eps_total	0.05	0.01, 0.05, 0.2	Design variable.	n/a (design)
level2_footprint_mult	3.0	2.0, 3.0, 6.0	Assumption (only used when needed).	low
level2_time_mult	2.0	1.5, 2.0, 3.0	Assumption (only used when needed).	low
model_A	1.0	0.5, 1.0, 2.0	Fit uncertainty: 9 points from Stim; exponent fixed to 1.	medium
model_pth	1.0	0.8, 1.0, 1.25	Fit uncertainty (Stim threshold ~0.7-1% for this noise model).	medium

Assumptions table. 'Confidence' is our judgement of how well the constant is pinned down; constants marked low are order-of-magnitude. The '+9.2' additive T-count of an earlier draft could not be verified and is not used.

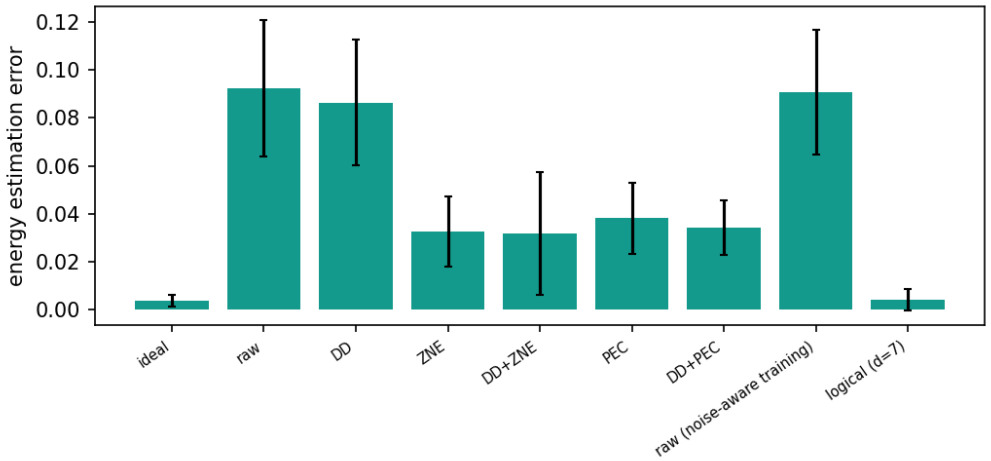
Not modelled: classical decoding latency and reaction time, factory idle errors, non-uniform layouts, measurement-based rotations injected without synthesis (analog-rotation architectures), and correlated physical noise. Estimates are order-of-magnitude tools for scaling arguments, not hardware quotes.

7 (b) Error-mitigation ablation

Instances are QUBOs from random historical dates; each configuration is run with finite shots (2048) on every instance with repeats. *est_err* = |estimated – ideal energy| (estimation accuracy); *gap_norm* = decoded-solution gap normalised by (mean – optimum) (0 = optimal); *p_opt_model* = exact probability of sampling the optimum under the noise model; *overhead* = sampling-variance multiplier of the mitigation.

config	est_err	est_err_sd	gap_norm	hit_opt	p_opt_model	N99	overhead	q_eff
ideal	0.00368	0.00254	0.00125	0.104	7.8e-05	5.72e+04	1	0
raw	0.0924	0.0283	0.00175	0.104	5.35e-05	8.62e+04	1	0.0464
DD	0.0864	0.0263	0.00164	0.104	5.43e-05	8.47e+04	1	0.0444
ZNE	0.0326	0.0147	0.00184	0.0833	5.35e-05	8.62e+04	57	0.0464
DD+ZNE	0.0318	0.0256	0.00171	0.115	5.43e-05	8.47e+04	57	0.0444
PEC	0.0381	0.0148	0.00158	0.135	5.35e-05	8.62e+04	24.3	0.0464
DD+PEC	0.0342	0.0115	0.00161	0.0833	5.43e-05	8.47e+04	20.9	0.0444
raw (noise-aware training)	0.0909	0.026	0.00164	0.104	5.33e-05	8.67e+04	1	0.0464
logical (d=7)	0.00424	0.00448	0.00135	0.135	7.8e-05	5.72e+04	1	3.04e-05

Ablation (averages over instances x repeats; N99 = median shots to see the optimum with 99% confidence under the model).



Energy-estimation error by configuration (mean ± sd over runs).

Findings. Raw noise gives est_err 0.092; the best physical-noise mitigation is *DD+ZNE* (0.032, overhead 57.0x); the logical-qubit mode reaches 0.004, indistinguishable from ideal (0.004, finite-shot floor). Mitigation reduces estimation bias but barely changes the probability of finding the optimum, which is set by q·n (theory P1): decoding already uses classical evaluation of every sample. ZNE pays a variance factor of 19 per Richardson extrapolation (P4) and PEC pays γ^2 (P3); both overheads grow exponentially in n·q.

8 (c) Out-of-sample results and statistical significance

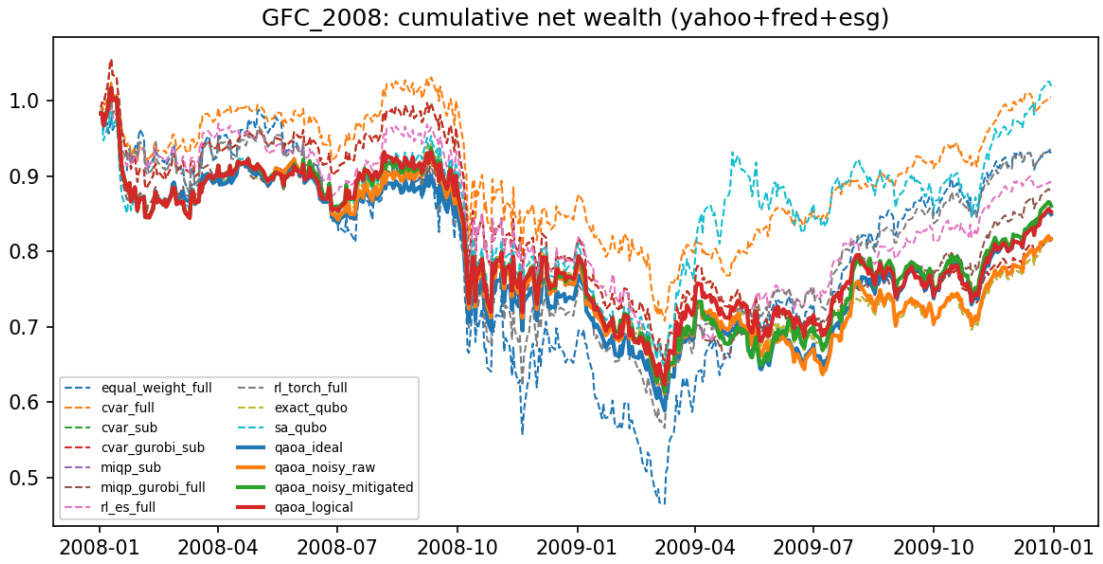
Walk-forward: at each rebalance the instance is built from past data only (unit-tested: perturbing the future does not change earlier decisions), solved, traded with 10.0 bp costs and held. Test windows: GFC 2008-01-01 to 2009-12-31 and COVID 2019-07-01 to 2020-12-31. Significance: paired stationary-bootstrap test of the Sharpe-ratio difference (mean block 10, 1000 resamples), Wilcoxon signed-rank on per-period returns, Holm correction over all quantum-vs-classical pairs.

Hyper-parameters: GFC_2008: tuned on pre-test data only {'lambda0': 6.0, 'kappa_vix': 0.0, 'esg_tilt': 0.002, 'tc_quad': 0.02}; COVID_2020: tuned on pre-test data only {'lambda0': 10.0, 'kappa_vix': 1.0, 'esg_tilt': 0.0, 'tc_quad': 0.005}.

index	ann_return	ann_vol	sharpe	max_drawdown	avg_turnover	avg_seconds
equal_weight_full	-0.0356	0.354	0.0741	-0.529	0.104	4.15e-05
cvar_full	0.00239	0.225	0.122	-0.313	0.18	0.0102
cvar_sub	-0.0974	0.249	-0.289	-0.384	0.66	0.00776
cvar_gurobi_sub	-0.0974	0.249	-0.289	-0.384	0.66	0.0165
miqp_sub	-0.099	0.255	-0.282	-0.389	0.701	0.00795
miqp_gurobi_full	-0.0618	0.236	-0.154	-0.383	0.291	0.0272
rl_es_full	-0.0562	0.244	-0.116	-0.389	0.151	0.000711
rl_torch_full	-0.0336	0.287	0.0233	-0.434	0.325	0.000256
exact_qubo	-0.099	0.255	-0.282	-0.389	0.701	0.000766
sa_qubo	0.00944	0.272	0.17	-0.346	0.825	0.0166
qaoa_ideal	-0.0785	0.26	-0.186	-0.421	0.735	6.89
qaoa_noisy_raw	-0.0964	0.26	-0.26	-0.398	0.768	4.38
qaoa_noisy_mitigated	-0.0726	0.262	-0.156	-0.396	0.697	3.72
qaoa_logical	-0.0768	0.256	-0.186	-0.387	0.701	3.72

GFC_2008: out-of-sample performance net of costs.

GFC_2008: 0 of 40 quantum-vs-classical Sharpe differences are significant after Holm correction (bootstrap); 0 by Wilcoxon. Smallest adjusted p = 1.000 (qaoa_ideal vs equal_weight_full, difference -0.26, 95% CI [-1.13, +0.50]).



GFC_2008: cumulative net wealth of all methods.

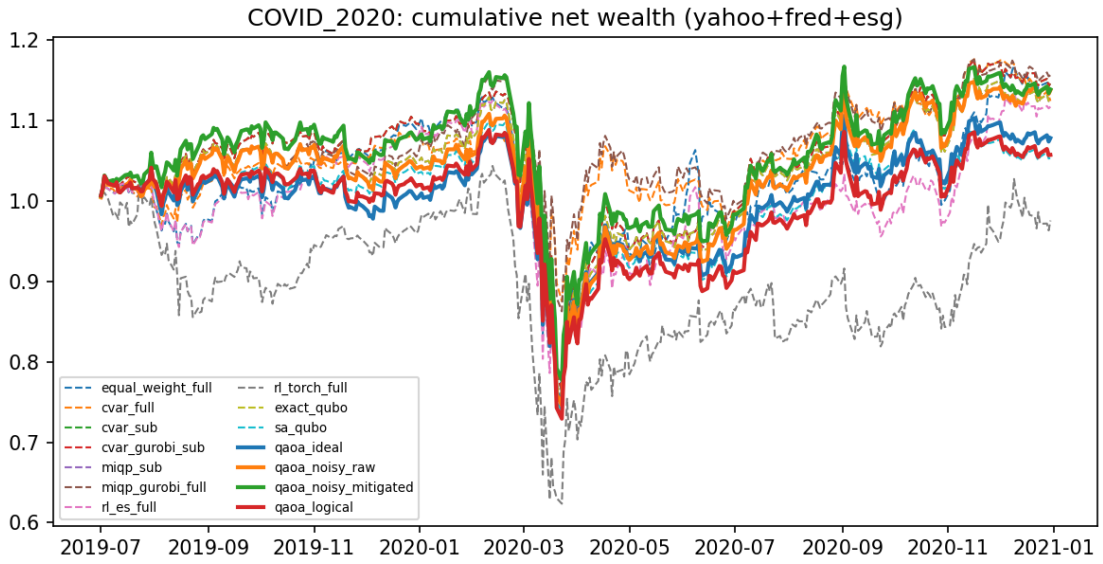
method	gap_norm	approx_ratio	optimum_hit_rate
exact_qubo	-3.3e-18	1	1
qaoa_ideal	0.00418	0.999	0.125
qaoa_logical	0.00342	0.999	0.208
qaoa_noisy_mitigated	0.00838	0.999	0.0417
qaoa_noisy_raw	0.00919	0.999	0.25
sa_qubo	0.00735	0.999	0.0417

GFC_2008: QUBO solution quality per rebalance (hit rate = fraction of rebalances where the exact optimum was returned).

index	ann_return	ann_vol	sharpe	max_drawdown	avg_turnover	avg_seconds
equal_weight_full	0.0948	0.282	0.463	-0.318	0.083	0
cvar_full	0.0807	0.226	0.456	-0.193	0.169	0.00572
cvar_sub	0.0949	0.261	0.479	-0.328	0.696	0.00554
cvar_gurobi_sub	0.0949	0.261	0.479	-0.328	0.696	0.0095
miqp_sub	0.0809	0.273	0.423	-0.33	0.743	0.00863
miqp_gurobi_full	0.101	0.256	0.504	-0.254	0.537	0.029
rl_es_full	0.0762	0.293	0.398	-0.336	0.131	0.000918
rl_torch_full	-0.0163	0.374	0.145	-0.402	0.695	0.00222
exact_qubo	0.0809	0.273	0.423	-0.33	0.743	0.000301
sa_qubo	0.0348	0.271	0.263	-0.329	0.846	0.00665
qaoa_ideal	0.0511	0.28	0.319	-0.309	0.881	6.26
qaoa_noisy_raw	0.0893	0.271	0.452	-0.327	0.73	4.63
qaoa_noisy_mitigated	0.0899	0.273	0.452	-0.33	0.744	1.95
qaoa_logical	0.0375	0.271	0.272	-0.33	0.777	2.42

COVID_2020: out-of-sample performance net of costs.

COVID_2020: 0 of 40 quantum-vs-classical Sharpe differences are significant after Holm correction (bootstrap); 0 by Wilcoxon. Smallest adjusted p = 1.000 (qaoa_ideal vs equal_weight_full, difference -0.14, 95% CI [-1.14, +0.72]).



COVID_2020: cumulative net wealth of all methods.

method	gap_norm	approx_ratio	optimum_hit_rate
exact_qubo	-1.44e-16	1	1
qaoa_ideal	0.00155	1	0
qaoa_logical	0.00173	1	0.211
qaoa_noisy_mitigated	0.00129	1	0.158
qaoa_noisy_raw	0.00143	1	0.158
sa_qubo	0.00251	1	0.0526

COVID_2020: QUBO solution quality per rebalance (hit rate = fraction of rebalances where the exact optimum was returned).

Statistical power. The median half-width of the 95% bootstrap interval for a Sharpe difference is 0.55 (GFC_2008), 0.50 (COVID_2020). Hence Sharpe differences smaller than roughly this size cannot be detected in a single 6-18 month window, and a non-significant result is *absence of evidence*, not evidence of equality. Equivalence (that quantum and classical solvers do not differ) would need the much stronger statement that the interval lies inside a pre-specified margin.

Soft vs hard budget. The exact QUBO optimum coincides with the hard-budget MIQP optimum in 100% of rebalances (GFC_2008); where it does, the QUBO penalty formulation loses nothing and differences between QUBO solvers in P&L are due to solver imperfection, not to the formulation.

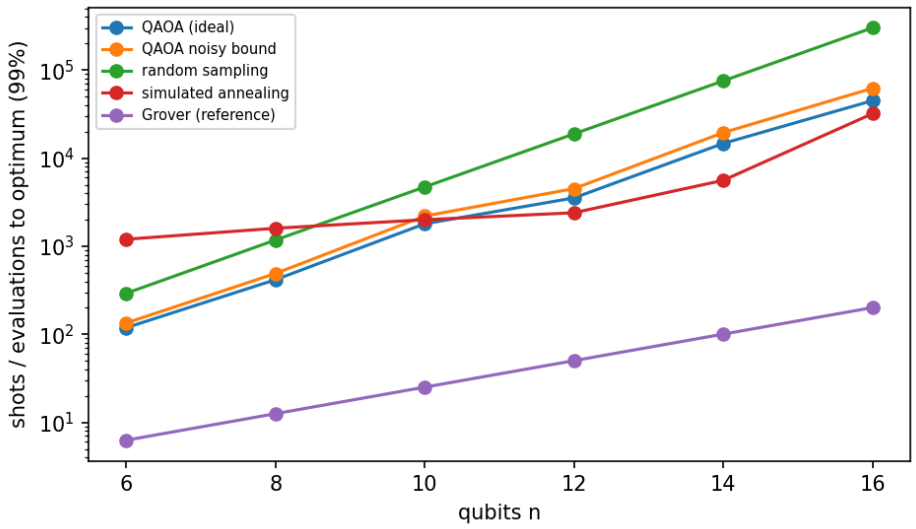
Interpretation guardrails. Differences among QUBO solvers reflect how often each returns the exact optimum on a 12-16 qubit problem, not investment skill; Sharpe differences over 6-18 months have wide

confidence intervals; the comparison against CVaR and RL conflates the objective (QUBO vs CVaR vs learned policy) with the solver.

9 Scaling study and theory

n	p_layers	p_opt_ideal	N99_qaoa	N99_qaoa_noisy_LB	N99_random	SA_evals_median	grover_queries
6	3	0.038	119	135	292	1.2e+03	6.28
8	3	0.011	417	491	1.18e+03	1.6e+03	12.6
10	3	0.00256	1.8e+03	2.2e+03	4.71e+03	2e+03	25.1
12	2	0.0013	3.54e+03	4.51e+03	1.89e+04	2.4e+03	50.3
14	3	0.000313	1.47e+04	1.95e+04	7.54e+04	5.6e+03	101
16	3	0.000102	4.5e+04	6.21e+04	301,802	3.2e+04	201

Shots (QAOA) or single-flip evaluations (SA) to reach the exact optimum with 99% confidence, vs qubits n (medians over instances).



Cost to reach the optimum vs n (log scale). Slopes are fitted exponents in $2^{slope \cdot n}$.

Fitted exponents (cost ~ $2^{s \cdot n}$): N99_qaoa: 0.85, N99_qaoa_noisy_LB: 0.87, N99_random: 1.00, SA_evals_median: 0.42, grover_queries: 0.50. **Caveats:** six sizes, n ≤ 16; p fixed by the adaptive rule; SA 'evaluations' are O(n)-cost single flips while a QAOA shot needs a full cost evaluation and a circuit execution, so the two axes are not directly comparable in wall-clock; SA uses restarts of 200n flips, which quantises its counts. This is trend evidence at toy size, not asymptotics.

Bounds proved for this noise model (docs/THEORY.md, each unit-tested)

- **P1** Success lower bound: $P_{noisy}(z) \geq P_{ideal}(z)(1-q)^n$ for every bitstring z.
- **P2** Energy estimation: $N \geq \Delta^2 \ln(2/\delta)/(2\varepsilon^2)$ shots (Hoeffding).
- **P3** PEC for independent flips: $\gamma = (1-2q)^{-n}$, shots $\propto \gamma^2$; with q = 0.03: $\gamma^2 = 7.2$ at n = 16, 2.4×10^5 at n = 100, 2.8×10^6 at n = 120.
- **P4** ZNE (scales 1,2,3): variance inflation 19 plus truncation bias.
- **P5** Random sampling needs $\ln(0.01)/\ln(1-2^{-n}) \approx 4.6\cdot2^n$ shots; Grover-type $(\pi/4)2^{n/2}$ is a reference line only.
- **P6** Noise budget: a shot-complexity advantage over a classical solver needing N_c evaluations requires $q \leq 1 - (p^*/P_{ideal}(z^*))^{1/n}$, which shrinks like 1/n - the quantitative reason logical qubits (q_L ~ 1e-5) are needed.

Short proofs and a data-derived noise budget

P1. The flip channel is a mixture: with probability $(1-q)^n$ no qubit flips and the output string equals the ideal one, so $P_{noisy}(z) = (1-q)^n P_{ideal}(z) + (\text{non-negative terms}) \geq (1-q)^n P_{ideal}(z)$. **P3.** One flip(q) channel acts on a qubit's distribution as the matrix [[1–q, q],[q, 1–q]] with inverse [[1–q, –q],[–q, 1–q]]/(1–2q), whose absolute entries sum (per column) to 1/(1–2q); for n qubits the quasi-probability norm is $\gamma = (1-2q)^{-n}$. Sampling signs from the inverse gives an unbiased estimator with range $\pm\gamma E_{max}$, hence Hoeffding gives $N = O(\gamma^2\Delta^2 \ln(1/\delta)/\varepsilon^2)$. **P6.** By P1 the shots to see z* with probability 0.99 are at most ln(0.01)/ln(1 –

$P_{ideal}(z^*)(1-q)^n$); requiring this to be $\leq N_c$ and solving for q gives the budget in the text.

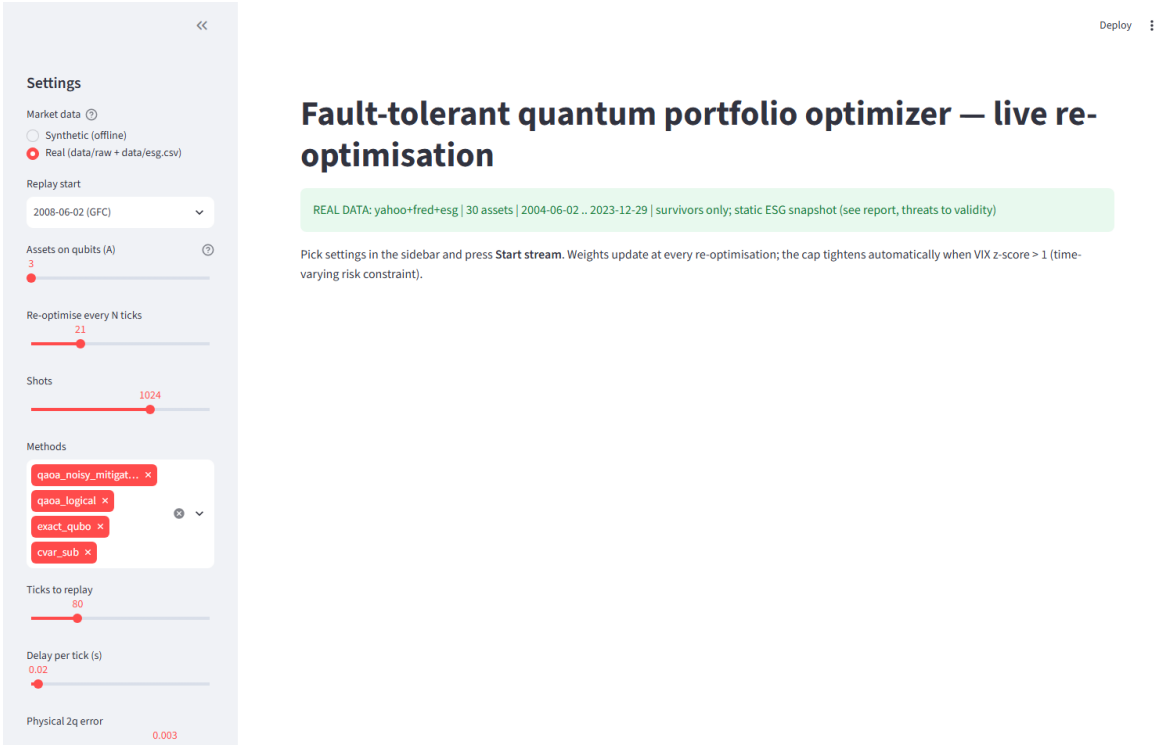
n	P_ideal(z*)	classical SA evals	max q for shot advantage	max q x n
6	0.038	1.2e+03	0.318	1.91
8	0.011	1.6e+03	0.154	1.23
10	0.00256	2e+03	0.0105	0.105
12	0.0013	2.4e+03	0	0
14	0.000313	5.6e+03	0	0
16	0.000102	3.2e+04	0	0

*Per-qubit flip probability q below which the guaranteed QAOA shot count (P1) stays under the median SA evaluation count, from our own instances. 0 means QAOA already needs more shots than SA with **no** noise. (Shots and SA flips are different units; this is the most generous reading for QAOA.)*

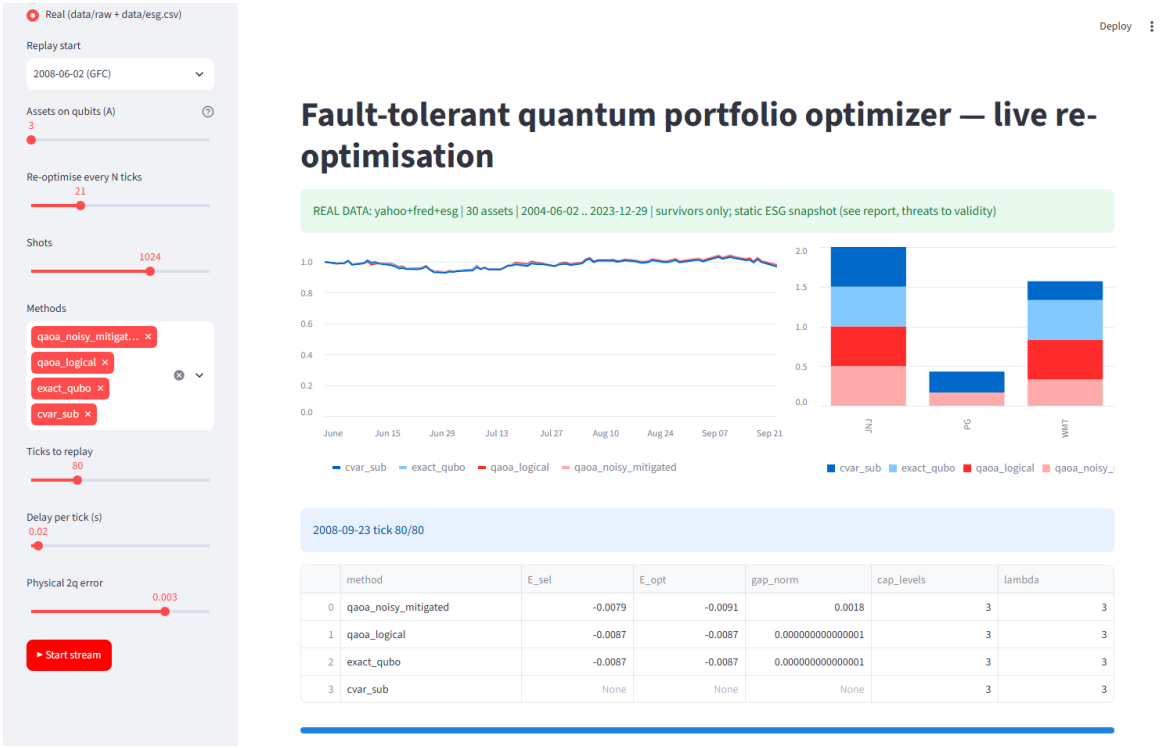
Not proven: any advantage of constant-depth QAOA over the best classical solvers; our scaling data are for n ≤ 16. Tang-style dequantization arguments for mean-variance structure are a further threat to any finance-specific advantage claim.

10 (d) Interactive dashboard: live re-optimisation under streaming data

`streamlit run dashboard/app.py` streams daily bars tick by tick; every N ticks it rebuilds the time-dependent QUBO from the data seen so far, re-solves it with each selected method (QAOA variants, exact, SA, MIQP, CVaR, equal weight), and updates P&L lines, target weights, the solver-gap table, the VIX-driven cap and risk aversion. A banner states whether the data are synthetic or real. Three data sources: synthetic, the real snapshot (a **streaming replay** of historical daily bars) and a **Live** option: `qpo/feed.py` defines the `next_bar()` hook, `YahooFeed` polls Yahoo Finance for daily bars newer than the last one and `catch_up()` extends the snapshot to the latest bar. Yahoo daily bars are end-of-day and quotes are about 15 minutes delayed, so «live» means daily-bar granularity plus delayed quotes; the Yahoo code was unit-tested with mocked payloads (the build sandbox has no Yahoo access) and falls back to the snapshot, saying so on screen, when offline.



Dashboard screenshot (real data).



Dashboard screenshot (real data).

10.1 Combined system: LSTM forecasts and fuzzy risk inside the optimiser

The root app (`streamlit run app.py`) joins the optimiser with the stock-prediction system. A pooled LSTM (price-only features, 8 outputs: cumulative log-returns at 1-7 and 21 days; trained offline, time-ordered split, train-only scalers, purged labels) supplies the expected returns μ of the QUBO (`mu_source = historical | lstm | blend`); a fuzzy risk score built from realised volatility, the VIX z-score and the forecast's out-of-sample confidence adds to the VIX term of λ_t (`risk_source = vix | vix+fuzzy`). The quantum part only optimises. `forecast_returns(tickers, as_of_date)` reads prices strictly before *as_of_date* and refuses any model trained on or after that date (walk-forward models, cut-offs at year ends; unit-tested). Transaction costs, ESG filter, caps and asset pre-selection are unchanged, and the same tuned configurations and causal rules are used, so the runs are paired.

method	historical	lstm	blend	blend+fuzzy
cvar_sub	-0.29	-0.29	-0.29	-0.29
equal_weight_full	0.074	0.074	0.074	0.074
exact_qubo	-0.28	-0.32	-0.21	-0.24
miqp_sub	-0.28	-0.32	-0.21	-0.24
qaoa_logical	-0.27	-0.1	-0.18	-0.19
qaoa_noisy_mitigated	-0.063	-0.19	-0.014	-0.08
sa_qubo	-0.1	-0.26	-0.11	-0.23

GFC_2008: out-of-sample Sharpe by solver and decision rule. equal_weight_full and cvar_sub do not use μ and are controls (identical across columns).

GFC 2008. 21-day forecast RMSE averaged over 23 rebalance dates: LSTM 0.1375, trailing-mean 0.1363, zero 0.1342; mean cross-sectional rank-free correlation with realised returns: LSTM -0.065, trailing-mean +0.015. Paired Sharpe differences against the historical-μ run that are significant after Holm correction: 0 of 21 (bootstrap), 0 of 21 (Wilcoxon). Quantum-vs-classical differences significant inside each variant: historical 0/10, lstm 0/10, blend 0/10, blend+fuzzy 0/10.

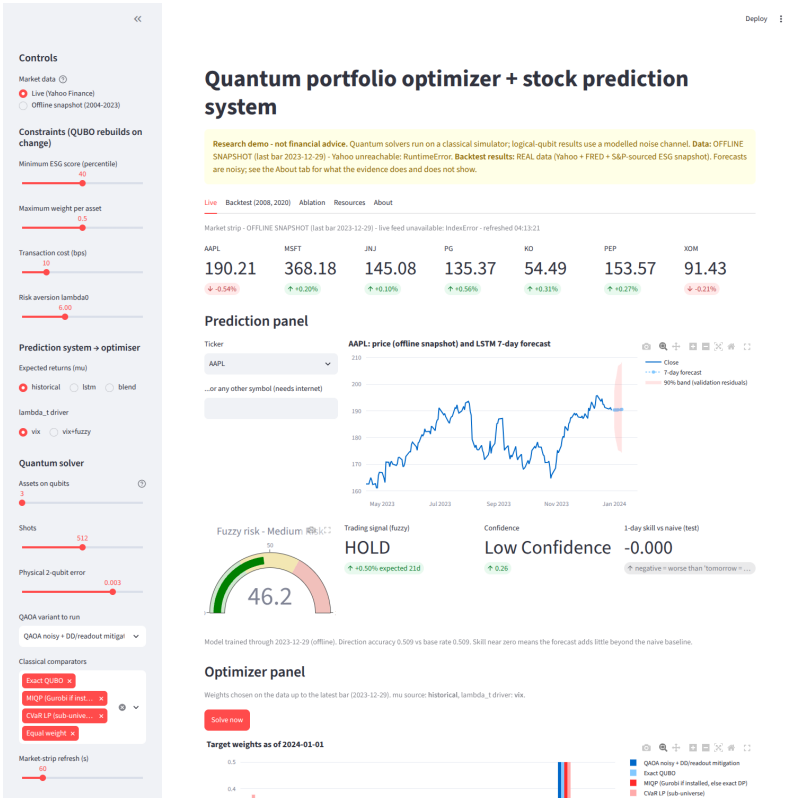
method	historical	lstm	blend	blend+fuzzy
cvar_sub	0.48	0.48	0.48	0.48
equal_weight_full	0.46	0.46	0.46	0.46
exact_qubo	0.42	0.37	0.45	0.47
miqp_sub	0.42	0.37	0.45	0.47
qaoa_logical	0.32	0.38	0.4	0.48
qaoa_noisy_mitigated	0.4	0.35	0.42	0.44
sa_qubo	0.41	0.5	0.58	0.3

COVID_2020: out-of-sample Sharpe by solver and decision rule. equal_weight_full and cvar_sub do not use μ and are controls (identical across columns).

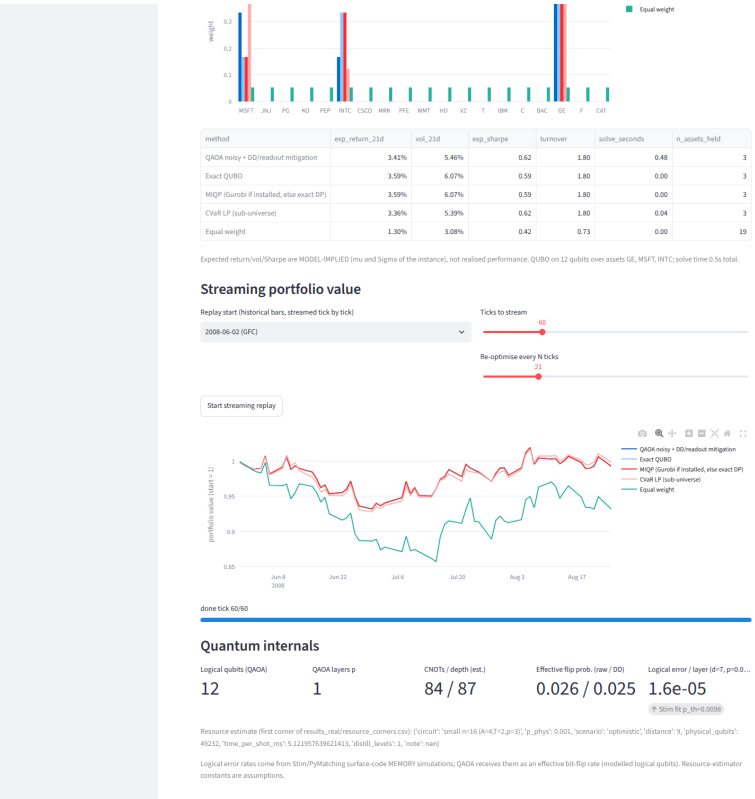
COVID 2020. 21-day forecast RMSE averaged over 18 rebalance dates: LSTM 0.0785, trailing-mean 0.0788, zero 0.0762; mean cross-sectional rank-free correlation with realised returns: LSTM +0.044, trailing-mean -0.075. Paired Sharpe differences against the historical-μ run that are significant after Holm correction: 0 of 21 (bootstrap), 0 of 21 (Wilcoxon). Quantum-vs-classical differences significant inside each variant: historical 0/10, lstm 0/10, blend 0/10, blend+fuzzy 0/10.

Reading. The forecaster does not beat the naive zero-return baseline out of sample, the portfolio differences between decision rules are small and not statistically distinguishable, and their signs flip between windows and solvers. The combination is a working pipeline, not evidence that predictions improve quantum portfolio selection.

Limitations of the combined system. (i) Forecast noise: near-zero skill at every horizon; the 90% band uses validation residual spread, not a calibrated predictive distribution. (ii) Delayed data: Yahoo daily bars/15-minute quotes, so nothing here is real-time trading. (iii) Small asset count: only 3-4 assets reach the qubits (greedy pre-selection is unchanged and drives most of the portfolio quality). (iv) The deployed model is trained on the shipped snapshot (to 2023-12-29) unless retrained with `--refresh`; the GA search used a deliberately small budget (6 × 3 evaluations) and is not used by the walk-forward backtest models. (v) Features are price-only (the snapshot has no volume), the fuzzy signal's volume rule therefore uses a neutral ratio in backtests.



Combined dashboard, Live tab (top): market strip, LSTM 7-day forecast band, fuzzy risk gauge, signal.



Combined dashboard, Live tab (bottom): solver weights, streaming portfolio value, quantum internals.

11 Software verification

Each claim in this report that can be checked mechanically is covered by a unit test (run with `python -m pytest -q`). The tests are the reason the reported numbers can be trusted to be the numbers the code means to produce:

test	what it establishes
QUBO == direct objective	build_qubo energy equals the hand-written multi-period objective on random bitstrings
QUBO optimum == MIQP optimum	with a large budget penalty the exact QUBO minimum equals the exact hard-budget MIQP optimum (cross-validates both formulations)
MIQP quadratic form	(H, c, const) given to Gurobi reproduces the objective for any integer levels, T = 1..3
QAOA normalised / improves	probabilities sum to 1; trained energy below uniform sampling
Flip channel inverse	apply_inverse_flip_channel exactly undoes apply_flip_channel
PEC exact / unbiased; ZNE reduces bias	infinite-shot PEC equals the ideal energy; sampled PEC is unbiased; ZNE error below raw
Theory P1, P2, P4, P3 values	P1 holds for every bitstring; Hoeffding and Richardson constants; gamma^2 reference values (7.24, 2.37e5)
No look-ahead	changing future returns leaves earlier backtest decisions unchanged
Real-data loaders (mocked)	all yfinance layouts, FRED old/new headers, no back-fill, truncated series dropped, late tickers dropped, pseudo-ESG refused
Ising / dimod conversions	Ising and dict energies equal QUBO energies; Ising phase layer equals the simulator's up to a global phase
Stim suppression; Gurobi == HiGHS; Qiskit == simulator; PyTorch agent	run only where stim / gurobipy / qiskit / torch are installed (skipped otherwise)

Which of the optional tests actually ran for this report is recorded in meta.json (`packages`) and in the status note shipped with the code.

12 Discussion: what a quantum advantage would require

Combining the pieces: (i) P6 says the tolerable per-qubit noise for a shot-complexity advantage decays like 1/n, so any advantage at scale is a logical-qubit statement; (ii) section 6 shows that a logical-qubit execution of 16 qubits needs about 75,576 physical qubits and 0.33 s per shot (about 11 minutes for 2048 shots), and 120 qubits need 58 s per shot (33 hours for 2048 shots), against microseconds for classical exhaustive search at n = 16; (iii) the fitted cost exponent of QAOA (0.85) grows faster than that of simulated annealing (0.42) on our instances, which limits what can be extrapolated; (iv) portfolio problems with realistic constraints (cardinality, round lots, sector limits) have far larger discrete search spaces than our 12-16 qubit encodings, but also much stronger classical solvers (branch-and-bound with convex relaxations, as in the Gurobi MIQP baseline) whose scaling we have not mapped. A credible advantage study would therefore need: instances with n >= 40 where exact methods fail, hardness-calibrated classical baselines (MIQP at its limits, tuned tabu/SA), QAOA depth scaling, and noise that is correlated and hardware-measured - none of which a 16-qubit statevector study can provide. What this project provides is the machinery and a negative baseline.

13 Threats to validity and conclusion

- **Survivorship bias.** Every ticker is a survivor (Lehman, Bear Stearns, WaMu, Wachovia are absent); this flatters all methods in 2008. The extended universe adds names that fell sharply but survived (C, BAC, AIG, GE, F...), reducing but not removing the bias.
- **Static ESG look-ahead.** ESG scores are a single present-day snapshot applied to 2005-2023; rating agencies also disagree strongly. The hard ESG filter therefore uses information unavailable at the time.
- **Sustainalytics polarity.** Risk scores (lower = better) are converted to a cross-sectional percentile score; the threshold then excludes a fixed fraction of the universe by construction.
- **Adjusted prices and macro data.** Dividends are embedded in adjusted prices; FRED's ICE-BofA credit-spread series is truncated to three years and is dropped, never back-filled.
- **Logical qubits are modelled, not run.** Stim simulates a surface-code *memory* experiment; QAOA receives the fitted logical error rate only as an effective independent bit-flip probability, so no logical gate, lattice-surgery or magic-state process is simulated. Resource-estimator constants (T-count per rotation, distillation footprint, cycle time, routing overhead) are assumptions, swept in results_real/resource_* and shown as corners and a tornado chart, never fitted to data.
- **Model-based noise.** NISQ+ noise is a phenomenological flip channel; real devices have correlated and coherent noise.
- **Small n.** At n ≤ 16 exhaustive search is trivial; all conclusions about scaling are extrapolations from six sizes.
- **Statistics.** Short windows, many pairs (Holm-corrected), and hyper-parameters tuned on windows preceding each test window only; the GFC validation window is calm so it cannot teach crisis behaviour.
- **Untested components.** none: all optional baselines ran.

Conclusion. Rate at which each solver returned the exact QUBO optimum at rebalance time: qaoa_ideal 7%, qaoa_logical 21%, qaoa_noisy_mitigated 9%, qaoa_noisy_raw 21%, sa_qubo 5%. Quantum-vs-classical Sharpe differences significant after Holm correction: GFC_2008: 0 of 40; COVID_2020: 0 of 40. Fitted cost-to-optimum exponents: QAOA 0.85 vs simulated annealing 0.42. The theory shows that noise budgets for an advantage shrink as 1/n and that PEC costs explode with n·q, and the resource estimates show that fault-tolerant execution of a 16-qubit instance needs 10^4-10^5 physical qubits. The supported claim is therefore *rigorous bounds plus an empirical test of advantage criteria*; no quantum advantage is demonstrated.

Appendix A: reproduction

```
python scripts/fetch_data.py # internet; writes data/raw
python scripts/build_esg.py # writes data/esg.csv
python scripts/run_all.py --stage qec # needs stim, pymatching
python scripts/run_all.py --stage resources
python scripts/tune.py --real --target GFC_2008; python scripts/tune.py --real --target COVID_2020
```

```
python scripts/run_all.py --real --require-optional --stage backtest,ablation,scaling,summary
python scripts/screenshots.py --real --out results_real
python report/build_report.py --results results_real --out report/QPO_Technical_Report.pdf
```

Appendix B: full significance tables

quantum	classical	sharpe_diff	ci_low	ci_high	p_boot	p_boot_holm	p_wilcoxon	p_wilcoxon_holm
qaoa_ideal	equal_weight_full	-0.26	-1.13	0.502	0.516	1	0.944	1
qaoa_ideal	cvar_full	-0.308	-1.11	0.368	0.395	1	0.473	1
qaoa_ideal	cvar_sub	0.103	-0.279	0.446	0.579	1	0.67	1
qaoa_ideal	cvar_gurobi_sub	0.103	-0.279	0.446	0.579	1	0.67	1
qaoa_ideal	miqp_sub	0.0962	-0.179	0.421	0.536	1	0.91	1
qaoa_ideal	miqp_gurobi_full	-0.0322	-0.633	0.502	0.906	1	0.584	1
qaoa_ideal	rl_es_full	-0.07	-0.873	0.689	0.875	1	0.989	1
qaoa_ideal	rl_torch_full	-0.209	-0.942	0.441	0.513	1	0.726	1
qaoa_ideal	exact_qubo	0.0962	-0.179	0.421	0.536	1	0.91	1
qaoa_ideal	sa_qubo	-0.355	-0.963	0.215	0.259	1	0.322	1
qaoa_noisy_raw	equal_weight_full	-0.334	-1.23	0.467	0.399	1	1	1
qaoa_noisy_raw	cvar_full	-0.383	-1.17	0.285	0.295	1	0.317	1
qaoa_noisy_raw	cvar_sub	0.0283	-0.336	0.378	0.879	1	0.989	1
qaoa_noisy_raw	cvar_gurobi_sub	0.0283	-0.336	0.378	0.879	1	0.989	1
qaoa_noisy_raw	miqp_sub	0.0216	-0.198	0.312	0.871	1	0.322	1
qaoa_noisy_raw	miqp_gurobi_full	-0.107	-0.674	0.421	0.69	1	0.726	1
qaoa_noisy_raw	rl_es_full	-0.144	-0.887	0.589	0.737	1	0.989	1
qaoa_noisy_raw	rl_torch_full	-0.284	-1.02	0.353	0.387	1	0.726	1
qaoa_noisy_raw	exact_qubo	0.0216	-0.198	0.312	0.871	1	0.332	1
qaoa_noisy_raw	sa_qubo	-0.43	-0.996	0.116	0.139	1	0.128	1
qaoa_noisy_mitigated	equal_weight_full	-0.231	-1.14	0.65	0.571	1	0.877	1
qaoa_noisy_mitigated	cvar_full	-0.279	-1.03	0.338	0.403	1	0.546	1
qaoa_noisy_mitigated	cvar_sub	0.132	-0.252	0.488	0.445	1	0.584	1
qaoa_noisy_mitigated	cvar_gurobi_sub	0.132	-0.252	0.488	0.445	1	0.584	1
qaoa_noisy_mitigated	miqp_sub	0.126	-0.17	0.469	0.426	1	0.848	1
qaoa_noisy_mitigated	miqp_gurobi_full	-0.00278	-0.55	0.512	0.993	1	0.966	1
qaoa_noisy_mitigated	rl_es_full	-0.0405	-0.815	0.717	0.914	1	0.747	1
qaoa_noisy_mitigated	rl_torch_full	-0.18	-0.903	0.48	0.558	1	0.944	1
qaoa_noisy_mitigated	exact_qubo	0.126	-0.17	0.469	0.426	1	0.848	1
qaoa_noisy_mitigated	sa_qubo	-0.326	-1	0.366	0.366	1	0.833	1
qaoa_logical	equal_weight_full	-0.26	-1.15	0.541	0.497	1	0.9	1
qaoa_logical	cvar_full	-0.308	-1.06	0.296	0.356	1	0.39	1
qaoa_logical	cvar_sub	0.103	-0.237	0.431	0.525	1	0.456	1
qaoa_logical	cvar_gurobi_sub	0.103	-0.237	0.431	0.525	1	0.456	1
qaoa_logical	miqp_sub	0.0964	-0.104	0.34	0.385	1	0.872	1
qaoa_logical	miqp_gurobi_full	-0.032	-0.546	0.462	0.899	1	0.747	1
qaoa_logical	rl_es_full	-0.0697	-0.822	0.633	0.869	1	0.989	1
qaoa_logical	rl_torch_full	-0.209	-0.94	0.396	0.476	1	0.726	1
qaoa_logical	exact_qubo	0.0964	-0.104	0.34	0.385	1	0.879	1
qaoa_logical	sa_qubo	-0.355	-0.915	0.185	0.218	1	0.527	1

GFC_2008: every quantum-vs-classical pair (bootstrap and Wilcoxon p-values, raw and Holm-adjusted).

quantum	classical	sharpe_diff	ci_low	ci_high	p_boot	p_boot_holm	p_wilcoxon	p_wilcoxon_holm
qaoa_ideal	equal_weight_full	-0.144	-1.14	0.725	0.738	1	0.679	1
qaoa_ideal	cvar_full	-0.137	-0.754	0.7	0.699	1	0.891	1
qaoa_ideal	cvar_sub	-0.16	-0.762	0.239	0.54	1	0.352	1
qaoa_ideal	cvar_gurobi_sub	-0.16	-0.762	0.239	0.54	1	0.352	1
qaoa_ideal	miqp_sub	-0.104	-0.501	0.133	0.522	1	0.332	1
qaoa_ideal	miqp_gurobi_full	-0.185	-0.635	0.271	0.368	1	0.515	1
qaoa_ideal	rl_es_full	-0.0789	-1.01	0.713	0.85	1	0.829	1
qaoa_ideal	rl_torch_full	0.174	-0.734	1.26	0.713	1	0.922	1
qaoa_ideal	exact_qubo	-0.104	-0.501	0.133	0.522	1	0.326	1
qaoa_ideal	sa_qubo	0.056	-0.223	0.273	0.657	1	0.983	1
qaoa_noisy_raw	equal_weight_full	-0.0106	-0.91	1.02	0.982	1	0.922	1
qaoa_noisy_raw	cvar_full	-0.0039	-0.666	1.02	0.993	1	0.541	1
qaoa_noisy_raw	cvar_sub	-0.0269	-0.405	0.294	0.858	1	0.891	1
qaoa_noisy_raw	cvar_gurobi_sub	-0.0269	-0.405	0.294	0.858	1	0.891	1
qaoa_noisy_raw	miqp_sub	0.0293	-0.116	0.209	0.678	1	0.379	1
qaoa_noisy_raw	miqp_gurobi_full	-0.0521	-0.506	0.658	0.859	1	0.738	1
qaoa_noisy_raw	rl_es_full	0.0543	-0.811	1.01	0.911	1	0.891	1
qaoa_noisy_raw	rl_torch_full	0.307	-0.597	1.61	0.564	1	0.953	1
qaoa_noisy_raw	exact_qubo	0.0293	-0.116	0.209	0.678	1	0.427	1
qaoa_noisy_raw	sa_qubo	0.189	-0.0539	0.54	0.2	1	0.196	1
qaoa_noisy_mitigated	equal_weight_full	-0.0102	-0.95	1.05	0.979	1	0.984	1
qaoa_noisy_mitigated	cvar_full	-0.00345	-0.648	1.01	0.994	1	0.49	1
qaoa_noisy_mitigated	cvar_sub	-0.0265	-0.383	0.271	0.865	1	1	1
qaoa_noisy_mitigated	cvar_gurobi_sub	-0.0265	-0.383	0.271	0.865	1	1	1
qaoa_noisy_mitigated	miqp_sub	0.0298	-0.108	0.196	0.648	1	0.46	1
qaoa_noisy_mitigated	miqp_gurobi_full	-0.0517	-0.491	0.577	0.863	1	0.86	1
qaoa_noisy_mitigated	rl_es_full	0.0547	-0.851	1.01	0.905	1	0.891	1
qaoa_noisy_mitigated	rl_torch_full	0.308	-0.644	1.63	0.578	1	0.953	1
qaoa_noisy_mitigated	exact_qubo	0.0298	-0.108	0.196	0.648	1	0.46	1
qaoa_noisy_mitigated	sa_qubo	0.19	-0.0512	0.565	0.211	1	0.136	1
qaoa_logical	equal_weight_full	-0.19	-1.16	0.747	0.675	1	0.568	1
qaoa_logical	cvar_full	-0.183	-0.789	0.687	0.637	1	0.86	1
qaoa_logical	cvar_sub	-0.206	-0.634	0.0914	0.263	1	0.21	1
qaoa_logical	cvar_gurobi_sub	-0.206	-0.634	0.0914	0.263	1	0.21	1
qaoa_logical	miqp_sub	-0.15	-0.469	0.0209	0.171	1	0.381	1
qaoa_logical	miqp_gurobi_full	-0.232	-0.698	0.339	0.347	1	0.709	1
qaoa_logical	rl_es_full	-0.125	-1.04	0.747	0.774	1	0.709	1
qaoa_logical	rl_torch_full	0.128	-0.826	1.28	0.796	1	0.768	1
qaoa_logical	exact_qubo	-0.15	-0.469	0.0209	0.171	1	0.307	1
qaoa_logical	sa_qubo	0.00959	-0.236	0.254	0.922	1	1	1

COVID_2020: every quantum-vs-classical pair (bootstrap and Wilcoxon p-values, raw and Holm-adjusted).

Appendix C: resource sensitivity (one parameter at a time)

circuit	parameter	qubits_min	qubits_max	ms_min	ms_max
small n=16	p_phys	16440	282552	152	631
small n=16	synth	56760	75576	109	435
small n=16	routing_overhead	71984	86352	327	327
small n=16	n_factories	26976	202776	70.8	1.31e+03
small n=16	factory_footprint_d2	43176	145776	327	327
small n=16	t_factory_cycles_per_T	75576	75576	178	653
small n=16	cycle_time_us	75576	75576	163	653
small n=16	cnot_parallel_frac	75576	75576	327	327
small n=16	overlap_T_and_clifford	75576	75576	327	328
small n=16	eps_total	56760	97080	244	430
small n=16	level2_footprint_mult	75576	75576	327	327
small n=16	level2_time_mult	75576	75576	327	327
small n=16	model_A	56760	75576	283	327
small n=16	model_pth	56760	97080	283	370
large n=120	p_phys	52308	2057364	1.24e+04	113,220
large n=120	synth	233748	539604	1.01e+04	7.73e+04
large n=120	routing_overhead	486744	698184	5.8e+04	5.8e+04
large n=120	n_factories	304524	1377396	1.31e+04	254,055
large n=120	factory_footprint_d2	349092	952380	5.8e+04	5.8e+04
large n=120	t_factory_cycles_per_T	539604	539604	3.16e+04	115,981
large n=120	cycle_time_us	539604	539604	2.9e+04	115,981
large n=120	cnot_parallel_frac	539604	539604	5.8e+04	5.8e+04
large n=120	overlap_T_and_clifford	539604	539604	5.8e+04	5.8e+04
large n=120	eps_total	233748	647316	2.36e+04	7.09e+04
large n=120	level2_footprint_mult	412596	920628	5.8e+04	5.8e+04
large n=120	level2_time_mult	539604	539604	4.35e+04	8.7e+04
large n=120	model_A	539604	539604	5.8e+04	5.8e+04
large n=120	model_pth	441684	647316	5.25e+04	6.35e+04

Range of physical qubits and time per shot (ms) when each parameter is swept over the values in the assumptions table, others at baseline.

Seeds: Config.seed = 0. Versions: numpy 2.4.6, scipy 1.15.3, pandas 2.3.1, stim 1.16.0, pymatching 2.4.0, torch 2.14.1+cpu, gurobipy 13.0.3, yfinance 1.2.0, qiskit 2.6.1. Results folder: results_real.